

PACMANN for convection dominant flows

In this work, we consider the steady convection–diffusion–reaction boundary value problem: find a sufficiently smooth solution $u : \bar{\Omega} \rightarrow \mathbb{R}$ such that

$$-\varepsilon \Delta u + b \cdot \nabla u + cu = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega, \quad (0.1)$$

where $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, is a bounded domain with polyhedral Lipschitz boundary $\partial\Omega$, $\varepsilon > 0$ denotes the diffusion coefficient, $b \in [W^{1,\infty}(\Omega)]^d$ is the convection field, $c \in L^\infty(\Omega)$ is the reaction coefficient, $f \in L^2(\Omega)$ is a given source term, and $g \in H^{1/2}(\partial\Omega)$ prescribes the Dirichlet boundary data. For simplicity, we restrict ourselves to Dirichlet boundary conditions, although the extension to other boundary conditions is straightforward.

In the proposed PINN framework, the solution u is approximated by a neural network $\hat{u}(x, \theta)$. In contrast to standard PINN approaches with fixed interior collocation points, we employ the Point Adaptive Collocation Method for Artificial Neural Networks (PACMANN), in which the interior collocation points are iteratively relocated according to the gradient of the squared residual.

More precisely, the residual of the governing equation is defined by

$$r(x, \theta) = -\varepsilon \Delta \hat{u}(x, \theta) + b(x) \cdot \nabla \hat{u}(x, \theta) + c(x) \hat{u}(x, \theta) - f(x), \quad (0.2)$$

and the adaptive movement of collocation points is guided by the quantity $r(x, \theta)^2$, so that more points are placed in regions where the differential equation is most strongly violated.

1 Hard-Constrained Physics-Informed Neural Networks (PINNs)

Physics-informed neural networks (PINNs) approximate the solution of boundary value problems by minimizing the residual of the governing equation together with the boundary conditions. In this work, we combine PINNs with PACMANN, which adaptively relocates interior collocation points according to the gradient of the squared residual, thereby concentrating collocation points in regions where the differential equation is most strongly violated.

Let $u_{\mathcal{N}}(x)$ denote a multilayer perceptron approximating the exact solution $u(x)$. For $x \in \bar{\Omega}$, the network output is defined recursively by

$$\hat{y}_1 := x, \quad (1.1)$$

$$\hat{y}_i := \sigma_i \left(W_i \hat{y}_{i-1} + \hat{b}_i \right), \quad i = 2, 3, \dots, n_L, \quad (1.2)$$

$$u_{\mathcal{N}}(x) := \hat{y}_{n_L}. \quad (1.3)$$

Here, W_i and $\hat{b}_i$ denote the weights and biases, respectively, and σ_i are activation functions. During the training process, the neural network parameters are optimized by minimizing a loss

functional

$$\mathcal{L} : \mathbb{R}^{d_p} \rightarrow \mathbb{R},$$

which maps the parameter vector to a non-negative real number measuring the approximation error.

In the PACMANN framework, the training problem is formulated as a min-max optimization problem

$$p^* \in \arg \min_{p \in \mathbb{R}^{d_p}} \left[\alpha_I \mathcal{L}_I + \alpha_D \mathcal{L}_D + \alpha_r \max_{X_r \subset \Omega} \mathcal{L}_r(X_r) \right], \quad (1.4)$$

where $X_r = \{x_r^i\}_{i=1}^{N_r}$ denotes the set of interior collocation points.

1.1 Examples of Loss Functionals

To ensure that neural networks approximate the solution of the boundary value problem, a loss functional must measure the discrepancy between the neural network prediction and the governing equations together with the boundary conditions.

Let

$$X_r = \{x_r^i\}_{i=1}^{N_r} \subset \Omega$$

denote the set of interior collocation points, and let

$$x_D^i \in \Gamma_D (= \partial\Omega), \quad i = 1, 2, \dots, N_D,$$

denote collocation points along the Dirichlet boundary. In PACMANN, the interior collocation points are not fixed but are iteratively relocated during training toward regions where the squared residual of the governing equation is large.

The loss functional is defined as

$$\mathcal{L} = \alpha_I \mathcal{L}_I + \alpha_D \mathcal{L}_D, \quad (5)$$

where $\alpha_I, \alpha_D \in \mathbb{R}$ are non-negative user-chosen constants. For convection-diffusion-reaction equations we define

$$\mathcal{L}_I(u_N) := \frac{|\Omega|}{N_r} \sum_{i=1}^{N_r} r(x_r^i, \theta)^2, \quad (1.5)$$

where the residual is given by

$$r(x, \theta) = -\varepsilon \Delta u_N(x) + b(x) \cdot \nabla u_N(x) + c(x) u_N(x) - f(x),$$

for $x \in \Omega$. The Dirichlet boundary loss is defined as

$$\mathcal{L}_D(u_N) := \frac{|\Gamma_D|}{N_D} \sum_{i=1}^{N_D} (u_N(x_D^i) - g_D(x_D^i))^2. \quad (1.6)$$

During the PACMANN procedure the interior collocation points x_r^i are updated iteratively according to the gradient of the squared residual. This corresponds to a gradient-ascent step of the form

$$x_r^{i+1} = x_r^i + s \nabla_x r(x_r^i, \theta)^2,$$

where $s > 0$ denotes a stepsize parameter controlling the magnitude of the update. Consequently, the collocation points move toward regions where the residual of the governing equation is large, improving the approximation quality of the neural network. In the above expressions we omit

the dependency of $u_{\mathcal{N}}$ on the network parameters p for simplicity. Furthermore, $|\Omega|$ denotes the d -dimensional measure of Ω and $|\Gamma_D|$ the $(d-1)$ -dimensional measure of Γ_D . For computing the derivatives of $u_{\mathcal{N}}$ in $\mathcal{L}_I$ it must be assumed that the activation functions are at least twice differentiable. If a Neumann boundary condition is present in (0.1), an additional term of the form

$$\frac{|\Gamma_N|}{N_N} \sum_{i=1}^{N_N} (\varepsilon \nabla u_{\mathcal{N}}(x_N^i) \cdot n(x_N^i) - g_N(x_N^i))^2$$

can be added using collocation points $x_N^i \in \Gamma_N$, $i = 1, 2, \dots, N_N$ along the Neumann boundary.

Now for the hp -variational loss functional, let $\mathcal{T}_h$ be a triangulation of $\bar{\Omega}$ into cells $K \in \mathcal{T}_h$. It is assumed that these cells are, depending on the dimension, either lines, quadrilaterals, or hexahedrons, and that they are the image of an affine or d -linear mapping

$$F_K : \hat{K} \rightarrow K$$

from the reference cell $\hat{K} := [-1, 1]^d$ to K .

For a given polynomial degree $p \in \mathbb{N}$, define the sets of test functions on the reference cells by

$$\begin{aligned} P_p([-1, 1]) &:= \{\varphi_j(\xi) : \varphi_j \in L_p([-1, 1]), j = 1, 2, \dots, p-1\}, \\ P_p([-1, 1]^2) &:= \{\varphi_j(\xi_1)\varphi_k(\xi_2) : \varphi_j, \varphi_k \in L_p([-1, 1]), j, k = 1, \dots, p-1\}, \\ P_p([-1, 1]^3) &:= \{\varphi_j(\xi_1)\varphi_k(\xi_2)\varphi_\ell(\xi_3) : \varphi_j, \varphi_k, \varphi_\ell \in L_p([-1, 1]), j, k, \ell = 1, \dots, p-1\}. \end{aligned} \quad (6)$$

Here

$$L_p([-1, 1]) := \{\varphi_{k+1}(\xi) - \varphi_{k-1}(\xi) : k = 1, \dots, p-1\},$$

where φ_k denotes the Legendre polynomial of degree k .

For each cell $K \in \mathcal{T}_h$, the corresponding set of test functions on the physical cell is defined by

$$P_p(K) := \{v : K \rightarrow \mathbb{R} \mid v(x) = \hat{v}(F_K^{-1}(x)), \hat{v} \in P_p([-1, 1]^d)\}.$$

The set of global polynomial test functions is then given by

$$P_p(\mathcal{T}_h) := \{v \in C(\bar{\Omega}) : v|_K \in P_p(K) \text{ for all } K \in \mathcal{T}_h, v|_{\partial\Omega} = 0\}.$$

By construction all test functions $v \in P_p(\mathcal{T}_h)$ have compact support and satisfy $v|_{\partial\Omega} = 0$. Let

$$X_r = \{x_r^i\}_{i=1}^{N_r} \subset \Omega$$

denote the set of interior collocation points used in the PACMANN framework. These points are iteratively relocated during training toward regions where the residual of the governing equation is large.

A variational loss functional for hp -PINNs combined with PACMANN is defined as

$$\mathcal{L}^{\text{hpvP}} = \alpha_I^{\text{hpvP}} \max_{X_r \subset \Omega} \mathcal{L}_I^{\text{hpvP}}(X_r) + \alpha_D^{\text{hpvP}} \mathcal{L}_D^{\text{st}}, \quad (7)$$

where $\alpha_I^{\text{hpvP}}, \alpha_D^{\text{hpvP}} \geq 0$ are user-chosen constants.

For convection–diffusion–reaction equations we define

$$\mathcal{L}_I^{\text{hpvP}}(X_r) := \sum_{K \in \mathcal{T}_h} \frac{1}{|P_p(K)|} \sum_{v \in P_p(K)} \left(\int_K (-\varepsilon \Delta u_{\mathcal{N}}(x) + b(x) \cdot \nabla u_{\mathcal{N}}(x) + c(x)u_{\mathcal{N}}(x) - f(x)) v(x) dx \right)^2, \quad (8)$$

where $|P_p(K)| := \dim(P_p(K))$.

1.2 Hard-Constrained PINNs

Instead of learning the Dirichlet boundary conditions through the loss functional, we impose them exactly by using a hard-constrained neural network ansatz

$$u_{\mathcal{N}}(x) = \tilde{g}_D(x) + h(x) \tilde{u}_{\mathcal{N}}(x),$$

where $\tilde{g}_D : \overline{\Omega} \rightarrow \mathbb{R}$ is a continuous extension of the boundary data g_D , and $h : \overline{\Omega} \rightarrow \mathbb{R}$ satisfies

$$h(x) = \begin{cases} 0, & x \in \Gamma_D, \\ > 0, & x \in \Omega. \end{cases}$$

Hence,

$$u_{\mathcal{N}}(x) = g_D(x) \quad \text{for } x \in \Gamma_D,$$

so the Dirichlet boundary conditions are satisfied exactly.

In the PACMANN framework, this is particularly convenient because the boundary collocation points remain fixed, while only the interior collocation points are adaptively relocated according to the gradient of the squared residual. Consequently, the boundary loss term $\mathcal{L}_D$ vanishes, and the training focuses on the interior residual.

2 Novel Loss Functionals for Physics-Informed Neural Networks

In convection-dominated convection–diffusion problems, the strong residual used in standard PINNs may lead to large values inside layers even for small solution errors.

Within the PACMANN framework, the interior collocation points

$$X_r = \{x_r^i\}_{i=1}^{N_r} \subset \Omega$$

are not fixed but are adaptively relocated during training toward regions where the squared residual is large. When hard-constrained PINNs are used (Sect. 2.3), the Dirichlet boundary conditions are satisfied exactly and the boundary loss term $\mathcal{L}_D^{\text{st}}$ vanishes.

2.1 Crosswind Loss Functional

To reduce spurious oscillations near layers, derivatives in the crosswind direction are penalized. Let the scaling function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$\Phi(x) = \begin{cases} \sqrt{x}, & x \geq 1, \\ 0.5(5x^2 - 3x^3), & \text{otherwise.} \end{cases} \quad (9)$$

Let $b(x) = (b_1(x), b_2(x))^T$ denote the convection field. In two spatial dimensions, a unit vector orthogonal to $b(x)$ is given by

$$b^\perp(x) = \begin{cases} \frac{(b_2(x), -b_1(x))^T}{\|b(x)\|_2}, & b(x) \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The crosswind-enhanced interior loss is defined by

$$\mathcal{L}_I^{\text{cw}}(X_r, \theta) = \mathcal{L}_I^{\text{st}}(X_r, \theta) + \frac{|\Omega|}{N_r} \sum_{i=1}^{N_r} \Phi(|b^\perp(x_r^i) \cdot \nabla u_{\mathcal{N}}(x_r^i, \theta)|), \quad (10)$$

where

$$\mathcal{L}_I^{\text{st}}(X_r, \theta) = \frac{|\Omega|}{N_r} \sum_{i=1}^{N_r} r(x_r^i, \theta)^2.$$

If Dirichlet boundary conditions are imposed weakly, the full crosswind loss functional is

$$\mathcal{L}^{\text{cw}}(X_r, \theta) = \alpha_I^{\text{cw}} \mathcal{L}_I^{\text{cw}}(X_r, \theta) + \alpha_D^{\text{cw}} \mathcal{L}_D^{\text{st}}(\theta). \quad (11)$$

If hard-constrained PINNs are used, then the Dirichlet boundary conditions are satisfied exactly and the boundary term vanishes. In this case, the PACMANN training problem takes the form

$$\theta^* \in \arg \min_{\theta} \left[\max_{X_r \subset \Omega} \mathcal{L}_I^{\text{cw}}(X_r, \theta) \right]. \quad (12)$$

In the numerical experiments with hard-constrained PINNs, we therefore set $\alpha_D^{\text{cw}} = 0$ and choose $\alpha_I^{\text{cw}} = 1$.